1. What is CPU bound task?

   => **CPU-bound task:**
   A **CPU-bound task** is a process that **spends most of its time using the CPU** for computation rather than waiting for I/O operations.

2. What is context switching?

   => **Context Switching:**
   Context switching is the process of **saving the state of one process and loading the state of another process** so the CPU can switch between them.

3. A web server handling multiple requests at the same time on multi-core CPU is an example of multiprogramming or multiprocessing?

   => **Answer: Multiprocessing**

   A web server handling multiple requests on a **multi-core CPU** is an example of **multiprocessing**, because multiple processes run **simultaneously on different CPU cores**.

4. Define dispatch latency.

   => **Dispatch Latency:**
   Dispatch latency is the **time taken by the operating system to switch the CPU from one process to another**.

5. What are the major activities of operating system?

   => **Major Activities of Operating System:**

   The operating system performs several important activities to manage the computer system:

   1. Process Management

   Creates, schedules, executes, and terminates processes.

   2. Memory Management

   Allocates and deallocates memory to processes and manages RAM efficiently.

   3. File System Management

   Handles creation, deletion, storage, and access of files and directories.

   4. Device Management

Controls input/output devices and manages device communication using drivers.

5. Secondary Storage Management

Manages disk space, storage allocation, and disk scheduling.

6. Security and Protection

Prevents unauthorized access and protects system resources and data.

7. User Interface

Provides interaction between user and system (CLI or GUI).

📌 **In short:**

The OS manages **processes, memory, files, devices, storage, security, and user interaction**.

6. What is the ready queue?

   => **Ready Queue:**
   The **ready queue** is a list of processes that are **ready to execute and waiting for CPU allocation**.

7. List the different state of a process.

   => **Process States:**
   New, Ready, Running, Waiting (Blocked), Terminated

8. What is non-preemptive scheduling?

   => **Non-Preemptive Scheduling:**
   It is a CPU scheduling method in which a process, once given the CPU, **runs until it finishes or enters waiting state without interruption**.